

CLINES: Clinical LLM-based Information Extraction and Structuring Agent

Zongxin Yang^{a,1}, Hongyi Yuan^{c,1}, Raheel Sayeed^{a,1}, Amelia Li Min Tan^a, Enci Cai^a, Mohammed Moro^a, Xiudi Li^d, Huaiyuan Ying^c, Nicholas Brown^f, Griffin Weber^{e,f}, Sheng Yu^c, Isaac Kohane^{a,*}, Tianxi Cai^{a,b,**}

^a*Department of Biomedical Informatics, Harvard Medical School, Boston, MA, USA*

^b*Harvard T.H. Chan School of Public Health, Boston, MA, USA*

^c*Department of Statistics and Data Science, Tsinghua University, Beijing, China*

^d*Division of Biostatistics, University of California Berkeley, Berkeley, CA, USA*

^e*Division of Interdisciplinary Medicine and Biotechnology, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA, USA*

^f*Harvard Medical School, Boston, MA, USA*

Abstract

Background. Clinical narratives in electronic health records (EHRs) contain essential diagnostic, therapeutic, and temporal information that is often missing from structured fields, leaving manual chart review as the *de facto* standard for high-quality labels, but slow, costly, and variable, thereby constraining accurate cohort construction for clinical trials, large-scale epidemiologic studies, and the development of robust machine-learning models.

Methods. We developed CLINES, a modular agentic pipeline that extracts and structures clinical concepts: semantic chunking of long notes; extraction by reasoning-capable large language models; assignment of attributes (assertion/experiencer, numerical values with SI units); normalization to the Unified Medical Language

*Corresponding author. Email: Isaac_Kohane@hms.harvard.edu.

**Corresponding author. Email: tcai@hsph.harvard.edu.

¹These authors contributed equally to this work.

System (UMLS); resolution of explicit and relative dates; and aggregation into an i2b2-style schema. Zero-shot evaluation was conducted on de-identified EHR: MIMIC-III notes, 4CE notes, and CORAL oncology reports (breast, pancreas). Comparators included rule/lexicon systems, transformer encoders, and single-prompt LLM baselines. Outcomes were F1 scores for entity extraction, assertion status, value&unit extraction, and date processing.

Finding. Across all datasets and evaluated tasks, CLINES led every baseline. F1 scores for entity / assertion / value&unit were: MIMIC-III 0.69 / 0.93 / 0.90; 4CE 0.87 / 0.88 / 0.79; CORAL–Breast 0.81 / 0.84 / 0.77; CORAL–Pancreas 0.85 / 0.87 / 0.90. Date processing was evaluated only on datasets with gold temporal annotations; F1 was 0.79 / 0.73 / 0.78 on 4CE / CORAL–Breast / CORAL–Pancreas. Gains over the strongest single-prompt LLM were +0.21–0.38 across tasks, and transformer encoders trailed by +0.28–0.68 F1 on entity extraction. Performance remained stable across note-length quantiles, while transformer baselines lost recall as notes lengthened.

Interpretation. CLINES translates narrative text from electronic health records into ontology-grounded, auditable, and schema-ready data, offering a practical route to scale chart-review-like extraction for cohort discovery and real-world evidence. CLINES is model agnostic—different open and closed models can be substituted to achieve specific cost, performance, and privacy goals. Future work aims to quantify inter-annotator agreements and explore adaptive feedback and domain-specific fine-tuning.

Keywords: artificial intelligence, large language models, clinical natural language processing, electronic health records, real-world evidence

1. Introduction

Clinical narratives in electronic health records (EHRs) are indispensable for understanding patient status, informing decision support, and enabling biomedical research. Narrative notes—spanning discharge summaries, progress notes, imaging and operative reports—capture observations, differential diagnoses, treatments, test results, and clinician reasoning that structured fields rarely encode. Expressions such as “intermittent chest pain worsening with exertion but relieved by rest” or “likely drug-induced hepatitis, possibly from isoniazid” capture nuance that structured fields miss; measurements like cardiac ejection fraction values or tumor size often do not fit neatly into predefined problem lists and billing codes yet are central to clinical interpretation. Interpretation is not just an important task of clinical interpretation for immediate clinician decisions. It also is the translation of this text to codified terminologies (e.g., CPT codes) or numerical representations (e.g., lab result tables) to drive tasks in epidemiology, automated decision support, reimbursement, and trial eligibility, among other downstream uses of electronic healthcare data.

Not only are the structured EHR entries incomplete or inaccurate, but conventional natural language processing methods—such as standard named entity recognition—often fail to capture the nuance of clinical reasoning, contextual qualifiers, or temporal relationships in notes¹. Even when structured codes or extracted entities are available, they may not reflect whether a condition was truly present, a medication was taken, or a disease state had recurred². Consequently, high-quality labels in practice still rely on chart review: investigators read notes to adjudicate diagnoses and verify supporting evidence (medications, symptoms, tests). Yet, chart review is slow, costly, and variable across annotators; double

abstraction is frequently required to ensure quality. This persistent reliance highlights the need for automated, scalable approaches that can more effectively extract, structure, and standardize clinical information consistently while leaving an auditable trace.

Classical approaches to clinical natural language processing have attempted to automate extraction and structuring of information from narrative notes but face important limitations. Rule- and lexicon-driven systems, such as cTAKES³, MetaMap⁴, and MedLEE⁵, map entities to controlled vocabularies (e.g., the Unified Medical Language System), using handcrafted rules and dictionaries for concept recognition, negation detection, and temporal normalization. While these pipelines established the foundation for clinical NLP, they struggle with shorthand, spelling variants and errors, as well as non-standard grammar common in clinical notes, and they require substantial manual engineering to adapt across institutions and use cases^{6,7}.

Building on these limitations of rule- and lexicon-driven systems, transformer-based large language models—such as GPT-4⁸ and Med-PaLM⁹—have reshaped NLP and show promise for clinical tasks including entity and relation extraction, summarization, and question answering¹⁰. Domain-scaled models, such as GatorTron¹¹ and Clinical-MobileBERT^{12,13}, report strong performance on benchmarks such as i2b2 concept extraction¹⁴, and instruction-tuned assistants can generalize in zero- and few-shot regimes¹⁵. Yet, important barriers remain for safe clinical deployment extraction: ontology grounding to UMLS/SNOMED CT is often imprecise with occasional hallucinated codes^{16,17}; context and output-length limits complicate reasoning across long clinical notes^{18,19}; and in real-world datasets (e.g., from the 4CE consortium), note lengths frequently exceed such limits, hin-

dering cross-sectional reasoning²⁰. As a result, single-pass prompting remains unreliable for chart-review-like decisions at scale.

To address these gaps, we present CLINES—a Clinical Language-model-based Information Extraction and Structuring agent that brings the consistency of chart review into an automated pipeline. Unlike prior systems limited by handcrafted rules or naive single-pass prompting, CLINES integrates several innovations: informed segmentation for processing long notes; reasoning-capable LLM extraction of entities and relations; attribution of clinical context (e.g., negation, experiencer, dosage, laboratory values); normalization to existing ontologies (e.g., UMLS); temporal alignment of events; and reconciliation into an i2b2-style schema to facilitate downstream integration. By uniting these components, CLINES represents a substantial step forward in clinical information extraction, with improvements over both classical NLP pipelines and baseline LLM prompting. Its improvements are validated using real-world EHR notes spanning multiple disease areas and institutions, underscoring both accuracy and generalizability. Equally important, CLINES is engineered to enable different generative models to be substituted within the workflow as dictated by user or institutional preferences regarding cost, property, performance and security.

2. Methods

2.1. CLINES architecture and modules

To leverage the superior performance of large language models within manageable context length, CLINES is designed as a four-step pipeline. Instead of an end-to-end approach—where LLMs often hallucinate or miss instructions in lengthy or complex notes—each key task is modularized and paired with carefully

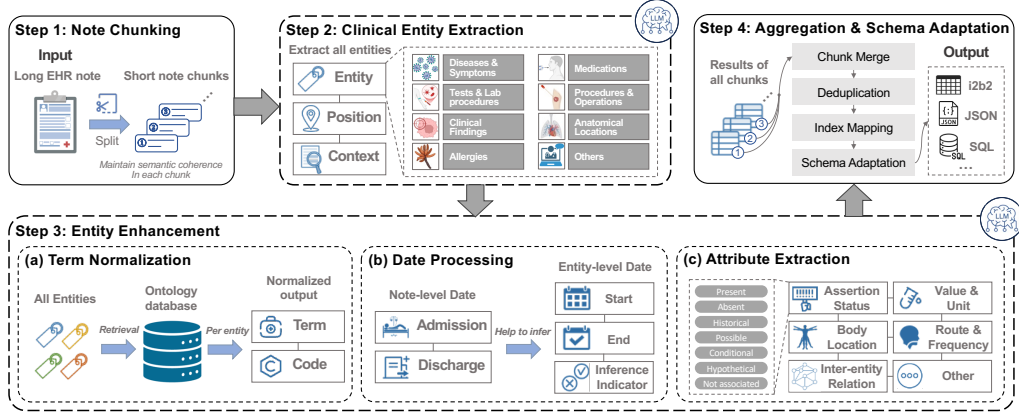


Figure 1: **Overview of CLINES.** Step 1, segments long clinical notes into semantically coherent chunks to constrain LLM’s context window and improve robustness. Step 2, from each chunk, extract clinical entities with broad coverage, including medications, diseases, and laboratory tests. Step 3, for each entity, uses local context to: (a) normalize to an ontology term with a unique code; (b) for event-type entities, extract or infer start and end dates, when imprecise; and (c) infer assertion status and capture clinically relevant modifiers, e.g., for medications: dose, unit, frequency, and route; for tests: numeric values (with units); for diseases: activity status and other pertinent attributes. Step 4, reconciles entity-level information across all chunks and exports results in user-specified formats (e.g., i2b2, JSON, SQL).

designed prompting strategies. The workflow is illustrated in Figure 1.

Step 1 – Note chunking. Input notes are tokenized with TikToken²¹ and segmented using SemChunk²² into semantically coherent fragments of roughly 768 tokens, preserving clinical meaning while fitting LLM context windows. Chunks are processed in parallel to increase throughput. Core metadata (e.g., admission or discharge dates) is retained in all chunks to support temporal alignment.

Step 2 – Clinical entity extraction. Each chunk is processed to identify relevant mentions (e.g., symptoms, findings, medications) in narrative order, with positions

and contexts preserved. Maintaining order captures clinician reasoning flow while isolating core concepts reduces redundancy and simplifies downstream relation, attribute, and temporal extraction.

Step 3 – Concept and attribute enrichment. **(a) Term normalization:** Extracted mentions are mapped to ontology concepts (e.g., UMLS) using LLM-assisted disambiguation and SapBERT-based retrieval, where SapBERT²³ provides biomedical embeddings; candidates are ranked by cosine similarity. **(b) Date processing:** Explicit dates are normalized to YYYY-MM-DD. Relative or implicit references (e.g., “three days post-admission”) are inferred using anchors such as admission and discharge timestamps; ambiguous dates are flagged as inferred. **(c) Attribute extraction:** For each normalized concept, the pipeline captures assertion status (present, absent, possible, historical), numerical values with inferred units, medication route and frequency, anatomical location, and relations between entities (e.g., infection “has symptom” presyncope, “managed by” rehydration). This enriched representation supports downstream phenotyping, predictive modeling, and real-world evidence generation^{24,25}.

Step 4 – Aggregation and schema adaptation. Outputs are reconciled across chunks through deduplication and cross-segment linking, unifying repeated mentions and preserving longitudinal continuity. Harmonized outputs are mapped to a standardized schema (e.g., i2b2) or a target schema specified by the use-case, then exported to CSV, JSON, or SQLite for downstream analysis. Module-level dataflow is summarized in Figure 2.

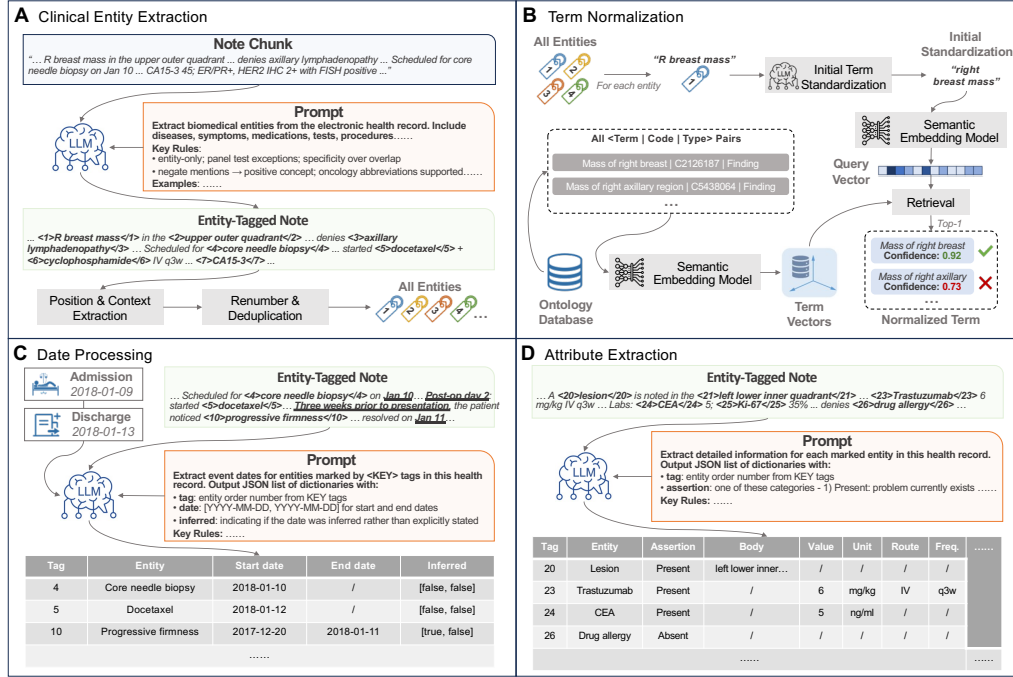


Figure 2: **CLINES LLM-based module workflow.** (A) Within each note chunk, the LLM tags clinical entities; post-processing yields an entity list with source offsets and a surrounding context window. (B) The LLM performs preliminary normalization, after which an embedding-based retriever queries the ontology database to return the nearest term, concept code, and semantic type. (C) Explicit dates/times are extracted from local context; when documentation is vague, admission/discharge metadata are used to resolve relative or fuzzy expressions to the most plausible calendar dates. (D) Context is used to capture modifiers (e.g., for medications: dose, unit, frequency, route; for tests: numeric values and units; for diseases: activity status), and to infer latent modifiers implied by panel tests (e.g., default units). All machine-inferred fields are flagged as “inferred” to facilitate audit and downstream use.

2.2. Implementation and LLM configuration

CLINES is model-agnostic; we employed three instruction-finetuned frontier LLMs with identical decoding settings: Llama-3.1-405B (temperature 0.6, max

output 8192 tokens; locally hosted on $8 \times$ H100 GPUs), GPT-4o (temperature 0.6, max output 8192), and o3-mini (medium reasoning effort, max output 32768). Notes were chunked to ≤ 768 tokens of raw text per request. With GPT-4o, mean prompt tokens were ≈ 8088.25 and mean completion tokens ≈ 2795.25 per chunk; with o3-mini, mean completion tokens increased to ≈ 24334.50 . GPT-4o and o3-mini were accessed via API (no local GPUs).

CLINES outputs were exported to the i2b2 schema, and a read-only web demonstrator populated with results from the credentialed-access CORAL dataset was deployed; access is described in the Data and Code availability section.

2.3. Study design and data

All corpora were de-identified and accessed under the respective data-use agreements and governance requirements of each dataset; no patient contact, intervention, or access to identifiable private information occurred.

MIMIC-III²⁶: from 2,065,096 critical-care notes, 964 notes were sampled and segmented into 2,000 paragraphs. Twenty-four paragraphs were manually annotated (one expert per paragraph), yielding 448 clinical phrases: 235 body-location mentions, 128 modifiers (e.g., assertion status), and 144 numerical values with units. Acronym–expansion pairs were treated as single phrases; narrative “purpose” attributes were excluded from quantitative analysis.

4CE²⁰: de-identified notes from patients with a history of COVID-19. Of 56 notes, 21 were annotated, producing 5,731 entity mentions with temporal information and attributes; mean document length 1,466 words (SD 636).

CORAL²⁷: 56 de-identified oncology reports (breast or pancreatic cancer). Twenty-nine notes were annotated and partitioned into two subsets: CORAL–Breast (13 documents, 3,900 entity records; mean 1,997 words) and CORAL–

Pancreas (16 documents, 4,717 entity records; mean 1,906 words). Across the annotated 4CE + CORAL subsets (14,348 entity records), 2,517 (17.5%) contained both a numerical value and a unit, and 5,680 (39.6%) carried explicit temporal information.

2.4. Annotation protocol

Five annotators followed a consensus protocol balancing accuracy and efficiency: (i) extract core entity spans while excluding modifiers to avoid redundancy; (ii) infer temporality and assertion status from surrounding context (e.g., “diagnosed with lung cancer 5 months ago,” “rule out pneumonia”); (iii) expand abbreviations and non-standard terms to full medical concepts, later normalized via SapBERT²³; (iv) infer missing units for numerical values. Annotation was supported by a custom React-based tool enabling interactive review, conflict resolution, and structured export. All annotators received unified training with shared examples; an independent annotator audited a sample for accuracy and consistency.

2.5. Baselines

To contextualize performance, we compared classical rule-/lexicon-driven pipelines and compact Transformer encoders, representing the two dominant traditions in clinical NLP.

Rule-/lexicon-based systems. cTAKES³ implements sectioning, sentence boundary detection, tokenization, POS tagging, morphological normalization, UMLS mapping, and ConText for negation, uncertainty, and experiencer; shallow parsing and dependency analysis link entities to attributes. MetaMap⁴ performs POS

tagging, variant generation, composite noun-phrase formation, and UMLS matching with heuristic scoring plus statistical word-sense disambiguation; it does not natively extract numerical values, units, or temporal relations.

Transformer encoders. Clinical-MobileBERT and Clinical-DistilBERT^{12,13} fine-tuned on i2b2-2010 concepts¹⁴ were included as parameter-efficient clinical NER baselines; evaluations were limited to entity extraction because they do not natively support ontology normalization, attribute extraction, or temporal reasoning. Model configurations and hyperparameters followed the authors’ implementations.

Single-prompt LLMs. Llama-3.1-405B and GPT-4o were also evaluated in a single-call setting using a compact prompt specifying the target schema and strict JSON output; notes were chunked only to satisfy context limits, and each chunk was processed once without retrieval, tools, cross-chunk aggregation, or multi-step reasoning.

2.6. Evaluation metrics and protocol

In the zero-shot setting (no training on the annotated test sets), we evaluated: entity detection (phrase-level precision, recall, F1 with exact-span match); attribute extraction (joint exact match of span and attribute for body location, assertion status, value/unit); date extraction and normalization (agreement with gold ISO dates; relative dates correct if the resolved absolute date matches the gold anchor). For Clinical-MobileBERT/Clinical-DistilBERT we report only NER; for cTAKES and MetaMap we report concept identification and assertion where available. End-to-end structuring, normalization, attribute extraction, and date handling are compared only across systems that produce those outputs.

3. Results

3.1. Benchmark overview

Across four corpora (MIMIC-III, 4CE, CORAL–Breast, CORAL–Pancreas) and four extraction tasks (entity, assertion status, value–unit as a single field, and date processing with start–end joint correctness), CLINES is benchmarked against rule/lexicon pipelines, transformer encoders, and single-prompt LLMs (Figure 3A–D). CLINES (o3-mini) ranks first on every available task–dataset combination. Relative to the best single-prompt LLM baseline, absolute F1 gains are +0.21–0.33 for entity extraction, +0.21–0.28 for assertion status, +0.27–0.38 for value–unit, and +0.26–0.33 for date processing. Versus transformer encoders (entity only), CLINES improves F1 by +0.28–0.68 across datasets. Non-LLM baselines trail on entities and do not produce value–unit or date outputs; among them, only cTAKES yields assertion labels.

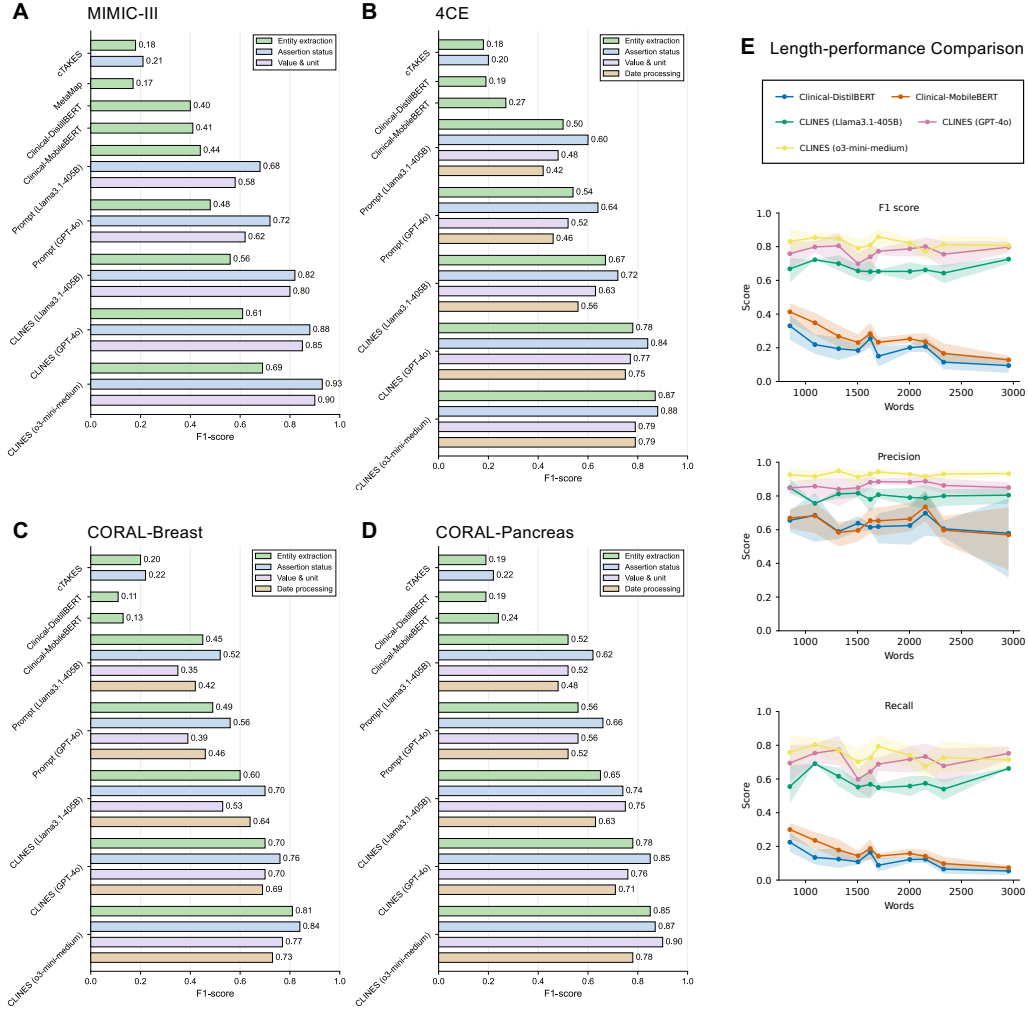


Figure 3: Performance of CLINES across datasets and tasks. (A–D) Results on MIMIC-III, 4CE, and CORAL—Breast/Pancreas comparing CLINES with traditional rule-based NLP, lightweight transformer-based baselines, and LLMs with single-step prompt engineering across six extraction tasks (treating value+unit as a single field and evaluating start+end date as joint correctness). The metric is F1 score. (E) CLINES versus transformer-based baselines as note length/complexity increases, reported as F1, precision, and recall. Notes are grouped into quantile-based length bins; markers indicate the bin-median word count; lines show bin-mean scores; shaded ribbons denote 95% bootstrap confidence intervals.

3.2. Across corpora and variants

The three CLINES variants maintain a consistent ordering—o3-mini > GPT-4o > Llama-3.1-405B—for entities, assertions, and value-unit, with all three delivering date extraction on 4CE and CORAL. Single-prompt LLMs outperform rule/lexicon and encoder baselines yet remain below CLINES on every task.

3.3. Note-length analysis

Across quantile-based length bins (~1–3k words), all CLINES variants show stable F1, precision, and recall (Figure 3E). Transformer-encoder baselines decrease with note length and complexity, driven primarily by falling recall, which remains substantially lower than CLINES across bins.

4. Discussion

4.1. Principal findings

The CLINES pipeline advances the automation of structuring clinical narratives, transforming narrative notes into standardized data formats that can be readily ingested into relational databases or analytical data warehouses. By design, CLINES bridges the longstanding divide between narrative clinical documentation and structured queryable formats, enabling scalable solutions for cohort discovery, real-world evidence generation, and clinical decision support. Critically, its robustness across diverse note types and lengths positions CLINES to unlock the vast reservoirs of unstructured text in EHRs. At scale, this capability can accelerate translational research, enhance population health monitoring, and power next-generation learning health systems by making rich narrative data systematically analyzable.

4.2. Backbone model considerations

Our findings reaffirm that backbone choice materially affects downstream performance. CLINES powered by o3-mini/GPT-4o consistently outperformed CLINES powered by Llama-3.1-405B, especially for contextual inference (e.g., implicit unit identification). This should not be interpreted as a direct head-to-head comparison — o3-mini/GPT-4o are proprietary models with broad pretraining on diverse text, whereas Llama-3.1-405B is open-source with more limited access to real-world clinical text. Performance disparities are therefore expected, underscoring the trade-off between accuracy and accessibility. A key advantage of CLINES is flexibility: the pipeline can be adapted to most LLMs so users can choose the backbone that aligns with performance, cost, and governance priorities. With the recent availability of GPT-OSS, CLINES can be straightforwardly updated to leverage such advances while maintaining accessibility.

4.3. Positioning versus existing approaches

Traditional clinical NLP—predominantly rule- or dictionary-driven—faces limited portability and high maintenance across institutions and note types; systems such as cTAKES and MetaMap emphasize lexicon/UMLS mapping and context heuristics but provide partial coverage for attributes or temporal relations. Compact transformer encoders fine-tuned for clinical NER are efficient for span detection yet generally stop short of normalization, assertion/date attribution, or cross-sentence temporal reasoning, and their recall degrades as notes lengthen. By contrast, single-prompt LLM baselines mitigate some gaps (handling heterogeneous phrasing and implicit cues) but remain brittle for schema-aligned outputs: they do not reliably perform controlled-vocabulary normalization and are more prone to hallucination without systematic designs. In CLINES, an agentic

orchestration—chunking, tool-guided extraction, normalization, and date reconciliation—delivers structured outputs with transparent intermediate artifacts and maintains robustness across diverse note types. This positions CLINES as a scalable and sustainable approach for structuring clinical narratives.

4.4. Case studies: robust clinical inference

De-identified 4CE and CORAL notes show four recurring capabilities: inferring units when measurements are underspecified; normalizing terse or site-specific shorthand to controlled vocabularies; recovering relations that mirror care processes; and handling nuanced assertion status and qualitative values. Examples include resolving “Lat 2.4,” “BE −30,” and “Glic 617” to lactate, base excess, and glucose with appropriate units and UMLS links; mapping abbreviations such as IOT, TI, RX, PCI, and EVAR to procedures or imaging; linking coronary artery disease to s/p PCI; chaining biliary obstruction → stent placement → infectious management in oncology follow-ups; and retaining Present / Absent / Possible / Conditional assertions and qualitative ranges (e.g., “mid-90s” oxygen saturation) in machine-actionable form. These vignettes show the system operating as a clinical reasoner rather than a string matcher, reducing manual normalization burden while increasing fidelity for phenotyping and decision support in heterogeneous, length-variable notes.

4.5. Dataset-level variation and generalization

Performance varies between datasets for a given model and task, reflecting different writing styles and note sizes across disease areas. The agentic, stepwise design limits error propagation: each module is independently validated, enabling interpretable error analysis. Semantic chunking is crucial for fitting long notes

into LLM context windows, allowing even 4k–8k-token models to operate effectively. UMLS term normalization addresses LLM imprecision with controlled vocabularies, unifying semantically equivalent phrases (e.g., “heart attack” and “myocardial infarction”) under consistent codes and enhancing interoperability.

4.6. Schema adaptability

CLINES adapts output to target schemas (e.g., i2b2 star schema, registries, decision-support formats) and exports JSON, CSV, or SQLite, supporting integration with institutional analytics pipelines and EHR-linked databases.

4.7. Implications

For cohort discovery, phenotyping, and real-world evidence, CLINES can replace or substantially reduce manual chart review for entities, assertions, values/units, and dates while leaving a verifiable audit trail. Sites can tailor prompts, dictionaries, and export schemas to local workflows without retraining large models.

4.8. Limitations

Several limitations remain. (i) Prompt engineering influences performance at module level; prompts were not dataset- or model-optimized. (ii) Comparisons among o3-mini/GPT-4o and Llama-3.1-405B are not apples-to-apples given architecture, data, and token-limit differences. (iii) Hallucination risk persists, especially for ambiguous references or rare abbreviations (e.g., “OD 2 gtt qd” misread as overdose rather than ophthalmic dosing). (iv) Gold annotations came from different annotators across datasets, introducing variability. Future work should quantify inter-annotator agreement and explore ensemble annotation strategies.

4.9. *Future work*

Future directions include domain-adaptive fine-tuning (e.g., oncology, pediatrics), adaptive feedback loops where clinician corrections refine prompts or parameters, and multilingual extensions to support non-English healthcare systems.

Contributors

Conceptualization & supervision: Tianxi Cai, Isaac Kohane. Methodology & software: Zongxin Yang, Hongyi Yuan, Raheel Sayeed, Amelia Li Min Tan (design); Zongxin Yang, Hongyi Yuan (implementation). Data curation & annotation: Enci Cai, Mohammed Moro, Xiudi Li, Zongxin Yang, Raheel Sayeed. Formal analysis: Zongxin Yang, Hongyi Yuan, Huaiyuan Ying. Data access & verification: Zongxin Yang, Raheel Sayeed. The i2b2 web demonstrator: Nicholas Brown, Griffin Weber. Writing—original draft: Zongxin Yang, Raheel Sayeed. Writing—review & editing: all authors. Accountability: all authors approved the final manuscript, and accept responsibility for its accuracy and integrity. The corresponding author had final responsibility for the decision to submit.

Data Sharing

Source clinical notes were accessed under data-use agreements and are not publicly shareable. MIMIC-III and CORAL can be requested via the PhysioNet credentialing process (credentialed access under the corresponding data use agreements and required training). 4CE data are available under consortium governance to qualified researchers upon request. Study-derived annotations (de-identified spans, attributes, and normalized concepts), prompt templates, and evaluation scripts will be made public upon publication. CLINES code will be released on

GitHub at publication. An interactive, read-only i2b2 web demonstrator populated with CORAL results is available at `celehs.hms.harvard.edu/i2b2_demo`.

Ethics Approval

Ethics approval/IRB review was not required because this study used only de-identified data accessed under the respective data-use agreements and governance requirements of each dataset, with no patient contact, intervention, or access to identifiable private information.

Declaration of Interests

We declare no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

References

- [1] Kohane IS, Aronow BJ, Avillach P, Beaulieu-Jones BK, Bellazzi R, Bradford RL, et al. What Every Reader Should Know About Studies Using Electronic Health Record Data but May Be Afraid to Ask. *Journal of Medical Internet Research*. 2021 Mar;23(3):e22219. Available from: <http://dx.doi.org/10.2196/22219>.
- [2] Kim MK, Rouphael C, McMichael J, Welch N, Dasarathy S. Challenges in and Opportunities for Electronic Health Record-Based Data Analysis and

Interpretation. Gut and Liver. 2023 Oct;18(2):201–208. Available from: <http://dx.doi.org/10.5009/gnl230272>.

- [3] Savova GK, Masanz JJ, Ogren PV, Zheng J, Sohn S, Kipper-Schuler KC, et al. Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES): architecture, component evaluation and applications. Journal of the American Medical Informatics Association. 2010 Sep;17(5):507–513. Available from: <http://dx.doi.org/10.1136/jamia.2009.001560>.
- [4] Aronson AR. Effective mapping of biomedical text to the UMLS Metathesaurus: the MetaMap program. In: Proceedings of the AMIA Symposium; 2001. p. 17.
- [5] Friedman C, Hripcsak G, DuMouchel W, Johnson SB, Clayton PD. Natural language processing in an operational clinical information system. Natural Language Engineering. 1995 Mar;1(1):83–108. Available from: <http://dx.doi.org/10.1017/S1351324900000061>.
- [6] Tayefi M, Ngo P, Chomutare T, Dalianis H, Salvi E, Budrionis A, et al. Challenges and opportunities beyond structured data in analysis of electronic health records. WIREs Computational Statistics. 2021 Feb;13(6). Available from: <http://dx.doi.org/10.1002/wics.1549>.
- [7] Pathak J, Kho AN, Denny JC. Electronic health records-driven phenotyping: challenges, recent advances, and perspectives. Journal of the American Medical Informatics Association. 2013 Dec;20(e2):e206–e211. Available from: <http://dx.doi.org/10.1136/amiajnl-2013-002428>.

- [8] OpenAI, Achiam J, Adler S, Agarwal S, Ahmad L, Akkaya I, et al.. GPT-4 Technical Report. arXiv; 2023. Available from: <https://arxiv.org/abs/2303.08774>.
- [9] Singhal K, Azizi S, Tu T, Mahdavi SS, Wei J, Chung HW, et al. Large language models encode clinical knowledge. *Nature*. 2023 Jul;620(7972):172–180. Available from: <http://dx.doi.org/10.1038/s41586-023-06291-2>.
- [10] Brown TB, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, et al.. Language Models are Few-Shot Learners. arXiv; 2020. Available from: <https://arxiv.org/abs/2005.14165>.
- [11] Yang X, Chen A, PourNejatian N, Shin HC, Smith KE, Parisien C, et al.. GatorTron: A Large Clinical Language Model to Unlock Patient Information from Unstructured Electronic Health Records. arXiv; 2022. Available from: <https://arxiv.org/abs/2203.03540>.
- [12] Rohanian O, Nouriborji M, Kouchaki S, Clifton DA. On the effectiveness of compact biomedical transformers. *Bioinformatics*. 2023 Feb;39(3). Available from: <http://dx.doi.org/10.1093/bioinformatics/btad103>.
- [13] Rohanian O, Nouriborji M, Jauncey H, Kouchaki S, Nooralahzadeh F, Clifton L, et al. Lightweight transformers for clinical natural language processing. *Natural Language Engineering*. 2024 Jan;30(5):887–914. Available from: <http://dx.doi.org/10.1017/S1351324923000542>.
- [14] Kohane IS, Churchill SE, Murphy SN. A translational engine at the national scale: informatics for integrating biology and the bedside. *Journal of*

- the American Medical Informatics Association. 2012 Mar;19(2):181–185.
Available from: <http://dx.doi.org/10.1136/amiajnl-2011-000492>.
- [15] Radford A, Narasimhan K, Salimans T, Sutskever I. Improving Language Understanding by Generative Pre-Training. OpenAI; 2018. Available from: https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf.
- [16] Karabacak M, Margetis K. Embracing Large Language Models for Medical Applications: Opportunities and Challenges. Cureus. 2023 May; Available from: <http://dx.doi.org/10.7759/cureus.39305>.
- [17] Rawte V, Sheth A, Das A. A Survey of Hallucination in Large Foundation Models. arXiv; 2023. Available from: <https://arxiv.org/abs/2309.05922>.
- [18] Zhao WX, Zhou K, Li J, Tang T, Wang X, Hou Y, et al. A Survey of Large Language Models. arXiv preprint arXiv:230318223. 2023; Available from: <https://arxiv.org/abs/2303.18223>.
- [19] Grattafiori A, Dubey A, Jauhri A, Pandey A, Kadian A, Al-Dahle A, et al. The Llama 3 Herd of Models. arXiv preprint arXiv:240721783. 2024; Available from: <https://arxiv.org/abs/2407.21783>.
- [20] Brat GA, Weber GM, Gehlenborg N, Avillach P, Palmer NP, Chiovato L, et al. International electronic health record-derived COVID-19 clinical course profiles: the 4CE consortium. npj Digital Medicine. 2020 Aug;3(1). Available from: <http://dx.doi.org/10.1038/s41746-020-00308-0>.

- [21] OpenAI. tiktoken. GitHub; 2024. GitHub repository, release 0.6.0 (published 2024-02-09). Available from: <https://github.com/openai/tiktoken>.
- [22] Isaacus-dev. semchunk. GitHub; 2024. GitHub repository, version 2.0.0 (released 2024-06-19). Available from: <https://github.com/isaacus-dev/semchunk>.
- [23] Liu F, Shareghi E, Meng Z, Basaldella M, Collier N. Self-alignment Pre-training for Biomedical Entity Representations. In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies; 2021. p. 4228–4238.
- [24] He T, Belouali A, Patricoski J, Lehmann H, Ball R, Anagnostou V, et al. Trends and opportunities in computable clinical phenotyping: A scoping review. *Journal of Biomedical Informatics*. 2023 Apr;140:104335. Available from: <http://dx.doi.org/10.1016/j.jbi.2023.104335>.
- [25] Hou J, Zhao R, Gronsbell J, Lin Y, Bonzel CL, Zeng Q, et al. Generate Analysis-Ready Data for Real-world Evidence: Tutorial for Harnessing Electronic Health Records With Advanced Informatic Technologies. *Journal of Medical Internet Research*. 2023 May;25:e45662. Available from: <http://dx.doi.org/10.2196/45662>.
- [26] Johnson AEW, Pollard TJ, Shen L, Lehman LwH, Feng M, Ghassemi M, et al. MIMIC-III, a freely accessible critical care database. *Scientific Data*. 2016 May;3(1). Available from: <http://dx.doi.org/10.1038/sdata.2016.35>.

- [27] Sushil M, Kennedy VE, Mandair D, Miao BY, Zack T, Butte AJ. CORAL: Expert-Curated Oncology Reports to Advance Language Model Inference. *NEJM AI*. 2024 Mar;1(4). Available from: <http://dx.doi.org/10.1056/aidbp2300110>.